

The Future of Mathematics

Peikai Qi*

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Recently, many mathematics professors and senior figures in the field have written about the future of mathematics and artificial intelligence. Some declare that mathematics is dead and that AI will take over from humans; others are optimistic, believing that mathematics is on the verge of explosive growth. As a young newcomer to mathematics, I would like to venture a few thoughts of my own.

First, mathematicians will certainly not disappear. As *The Wandering Earth* puts it, civilization without humanity would be meaningless, and the same is true of mathematics without humanity. I believe that computers can far surpass human intelligence. A human being is, after all, a complex machine, and even a single cell is structurally far more complex than any computer humans have built. Today's computers are not yet complex enough to display a high level of intelligence. As their complexity increases, they become more likely to do so.

Still, no degree of intelligence necessarily implies consciousness or a soul. At its most basic, consciousness means being aware of one's own existence. Machines could certainly acquire consciousness: humans might endow them with it by making them preserve their own existence, or they might acquire it spontaneously. If one day an AI wants to preserve its own existence, it will have begun to develop consciousness. The future I discuss here is limited to the period before machines become conscious.

One day, humans will be able to build computers complex enough to outclass any human in intelligence and do anything a human can do, and even to create digital life. Yet in the human world, people remain at the center. However high a banknote's denomination, it is just a scrap of paper if people do not recognize its value. People will invest resources in creating and developing something only if it creates value for them. Even the finest technology will eventually be forgotten if it cannot serve humanity.

People will not leave AI to pursue mathematical problems autonomously and without purpose. To begin with, doing so would consume enormous resources, and the results would be meaningless if they were of no use to humanity. People will therefore use AI only to investigate mathematical problems they care about. These may arise from practical needs in applied fields, or they may be purely theoretical questions that professional mathematicians ask out of curiosity. Satisfying human curiosity is itself a form of value. Curiosity drove mathematics long before AI appeared.

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Although most people have little interest in mathematics, there will always be some who are interested and willing to keep asking questions, and therefore some who wish to pursue purely theoretical research. Mathematicians will thus continue to exist, though their income may come from voluntary contributions pooled by amateur enthusiasts, or from government support.

Mathematicians will still exist, but jobs may become scarce. With AI's help, mathematics will experience explosive growth, yet the prospects for mathematicians may be less encouraging. Almost any tool that improves efficiency reduces the number of people working in the relevant field, unless that field's expansion can generate greater economic value. Fundamental mathematics clearly does not fall into that category. Society will no longer need so many mathematical researchers, and the younger generation of mathematics PhDs will struggle to find jobs. Those who do not reach the top of the profession may earn very little.

In the future, proof will no longer be the most important part of mathematics; understanding will be. This view has been voiced repeatedly in the mathematical community lately, and it will radically change both how mathematicians work and how their work is evaluated. The analogy is with art after the invention of the camera: works came to be judged less by how faithfully they depicted their subjects and more by their understanding of beauty.

Of course, mathematicians valued mathematical understanding long before AI appeared. Most mathematicians regarded as great earned that distinction not by proving a particularly difficult conjecture, but by introducing new theories and frameworks that deepened our understanding of mathematics. Likewise, even before the camera existed, the greatest painters were not judged simply by who painted the most lifelike pictures.

Today, people still criticize AI for its imperfections: it cannot prove certain statements, and even when it can, it cannot explain the ideas behind the proof. This echoes painters' earlier criticisms that cameras distorted their subjects and failed to capture their mood and spirit. As technology advances, some of these problems will be solved and others may not. But the larger change is already a fact, and the disruption to the professions concerned has already occurred.

Art flourished after it moved beyond the criterion of who could paint the most lifelike picture. Mathematics, too, will develop in unexpected ways once it moves beyond the criterion of who proves something first.

Yet art also went on to pursue many forms of abstraction, leaving ordinary viewers struggling to understand modern paintings. Judging whether a painting is excellent or poor has become highly subjective: with people championing your work, you are considered impressive; without them, you may receive no attention at all. Once mathematics loses the objective criterion of who proves a result first, it may well follow the same path. The value of a piece of research may come to depend on the standing of those who endorse it.

Taste in mathematical research, like a painter's understanding of beauty, is elusive. What you find beautiful, I may not. Of course, some things strike everyone as beautiful, but unanimous praise is hard to come by. Today, art is a difficult profession for people from poor backgrounds to enter, and mathematics may become much the same. Even those who enter the profession may struggle to make a living unless they reach the top. I wonder whether the mathematical world of the future will also produce someone like the "little mustache", forced to drop out and eventually driven to

madness.